

# RANJAN SARKAR

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PhD Research Scholar  
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 [LinkedIn](#)  
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 [GitHub](#)

## About

I am a PhD scholar at Department of Artificial Intelligence (AI), IIT Kharagpur. I am working in the [RHPI \(Receding Horizon and Predictive Intelligence\)](#) Lab under the supervision of [Dr. Prabhat Kumar Mishra](#). My research focuses on developing new robust RL algorithms with applications in LLM, Robotics, etc.

## Research Interest

Machine Learning • Optimization • Reinforcement Learning • State-Space Model

## Work Experience

2023 – 2024: Senior Business Analyst • **EXL Service**  
Gurgaon, India

- Worked on **BAU** operations.
- Responsible for data pipelining and analytics with **SQL** and **SAS**.

## Education

2025 – Present: Indian Institute of Technology Kharagpur  
PhD, Artificial Intelligence  
CGPA: 9.33/10

2021 – 2023: Indian Institute of Technology Kharagpur  
Master of Science (MSc), Mathematics  
CGPA: 8.76/10

2018 – 2021: Ramakrishna Mission Vidyamandira (University of Calcutta)  
Bachelor of Science (BSc), Mathematics  
CGPA: 9.15/10

2018: Raiganj Coronation High School (HS) • WBCHSE  
12<sup>th</sup> Standard, Science Stream  
Percentage: 88.6%

2016: Raiganj Coronation High School (HS) • WBBSE  
10<sup>th</sup> Standard  
Percentage: 94.57%

## Publications

Ranjan Sarkar, Prabhat K. Mishra, “[LQG Solution for POMDP without estimating states: A minimum variance approach](#)”, Indian Control Conference (ICC), 2027.

## Projects

### Master’s Thesis: Image Compression (2022-2023)

Supervisor: Prof. Swanand Khare, Department of Mathematics, IIT Kharagpur

- **Tensor CP decomposition** is implemented in a Python notebook.
- The **Alternative Least Square (ALS)** Optimization Method has been applied to get the low-rank approximation of the RGB image tensor.

## Teaching Assistant

AI31201 • Reinforcement Learning (Autumn 2026)

AI60209 • AI for Robot Autonomy (Autumn 2026)

AI61006 • AI for Cyber Physical Systems (Spring 2026)

Course Instructor - Dr. Prabhat Kumar Mishra

AI60211 • Algorithmic and Mathematical Foundations for AI (Autumn 2025)

Course Instructor - Dr. Adway Mitra

## Courses & Skills

- |                            |                                 |
|----------------------------|---------------------------------|
| • Linear Algebra           | • Advanced Numerical Techniques |
| • Probability & Statistics | • Convex Optimization           |
| • Real Analysis            | • Non-Linear Programming        |
| • Machine Learning         | • AI for Cyber Physical Systems |
| • Deep Learning            | • AI for Robot Autonomy         |
| • Artificial Intelligence  |                                 |

C, Python, LaTeX, Numpy, Pandas, Sympy, CasADi, PyTorch, TensorFlow, Scikit-Learn, Jupyter, etc.

## Awards & Achievements

1. Qualified **Joint CSIR-UGC NET** (June 2024) with **All India Rank (AIR) 83**.
2. **INSPIRE Scholarship** Recipient (2018-2023), by DST (Department of Science and Technology), Govt. of India - awarded for top 1% in WB government HS board examination (2018).

## Extra-curricular activities

Photography, Singing, Swimming, Table tennis.